# Experiment 4 - Migrate to Gradle

## Part A: Build and Run a Java Application with Maven

For this part, please refer to **Experiment 2: Working with Maven**. Follow the steps outlined in Experiment 2 to:

1. Create a Maven project (`HelloMaven`).

```
mvn archetype:generate -DgroupId=com.example -DartifactId=HelloMaven -
DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```

1. Navigate into the project directory.
2. Explore the project structure.

**Step 1 – Update pom.xml**

gedit pom.xml

```
<project xmlns="http://maven.apache.org/POM/4.0.0"

        xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"

        xsi:schemaLocation="http://maven.apache.org/POM/4.0.0

        https://maven.apache.org/xsd/maven-4.0.0.xsd">



    <modelVersion>4.0.0</modelVersion>



    <groupId>com.example</groupId>

    <artifactId>HelloMaven</artifactId>

    <version>1.0-SNAPSHOT</version>

    <packaging>jar</packaging>



    <name>HelloMaven</name>

    <url>http://maven.apache.org</url>
```

```
<properties>
    <maven.compiler.source>17</maven.compiler.source>
    <maven.compiler.target>17</maven.compiler.target>
    <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
    <junit.version>5.10.0</junit.version>
</properties>

<dependencies>

    <!-- JUnit 5 -->
    <dependency>
        <groupId>org.junit.jupiter</groupId>
        <artifactId>junit-jupiter</artifactId>
        <version>${junit.version}</version>
        <scope>test</scope>
    </dependency>

</dependencies>

<build>
    <plugins>

        <!-- Compiler Plugin -->
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
```

```
            <artifactId>maven-compiler-plugin</artifactId>
            <version>3.11.0</version>
        </plugin>

        <!-- Surefire Plugin for JUnit 5 -->
        <plugin>
            <groupId>org.apache.maven.plugins</groupId>
            <artifactId>maven-surefire-plugin</artifactId>
            <version>3.2.5</version>
        </plugin>

    </plugins>
  </build>

</project>
```

**Step 2 – Update AppTest.java**

gedit src/test/java/com/example/AppTest.java

**and replace the contents with**

```
package com.example;

import org.junit.jupiter.api.Test;
import static org.junit.jupiter.api.Assertions.assertTrue;

class AppTest {
```

```
    @Test
    void shouldAnswerWithTrue() {
        assertTrue(true);
    }
}
```

**3.Build the project using**

mvn clean test

`mvn package`.

```
skit@MANJUSHREENAVEEN: ~    skit@MANJUSHREENAVEEN: ·
ping: google.com: Temporary failure in name resolution
skit@MANJUSHREENAVEEN:~/HelloMaven$ mvn clean test
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:HelloMaven >-----------------------
[INFO] Building HelloMaven 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ HelloMaven ---
[INFO] Deleting /home/skit/HelloMaven/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ HelloMave
n ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /home/skit/HelloMaven/src/main/resou
rces
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ HelloMaven
---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 1 source file with javac [debug target 17] to target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ H
elloMaven ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /home/skit/HelloMaven/src/test/resou
rces
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:testCompile (default-testCompile) @ Hel
loMaven ---
```

```
skit@MANJUSHREENAVEEN: ~    skit@MANJUSHREENAVEEN: ·
[INFO]
[INFO] -------------------------------------------------------
[INFO]  T E S T S
[INFO] -------------------------------------------------------
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.157 s
-- in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  28.709 s
[INFO] Finished at: 2026-02-24T05:58:43Z
[INFO] ------------------------------------------------------------------------
skit@MANJUSHREENAVEEN:~/HelloMaven$ mvn package
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:HelloMaven >-----------------------
[INFO] Building HelloMaven 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ HelloMave
n ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /home/skit/HelloMaven/src/main/resou
rces
```

**4.Run the application using the `java -cp`**

mvn clean package -DskipTests

```
skit@MANJUSHREENAVEEN:~/HelloMaven$ mvn clean package -DskipTests
[INFO] Scanning for projects...
[INFO]
[INFO] -----------------------< com.example:HelloMaven >-----------------------
[INFO] Building HelloMaven 1.0-SNAPSHOT
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
[INFO] --- maven-clean-plugin:2.5:clean (default-clean) @ HelloMaven ---
[INFO] Deleting /home/skit/HelloMaven/target
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ HelloMave
n ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /home/skit/HelloMaven/src/main/resou
rces
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:compile (default-compile) @ HelloMaven
---
[INFO] Changes detected - recompiling the module! :source
[INFO] Compiling 1 source file with javac [debug target 17] to target/classes
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ H
elloMaven ---
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /home/skit/HelloMaven/src/test/resou
rces
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:testCompile (default-testCompile) @ Hel
loMaven ---
[INFO] Changes detected - recompiling the module! :dependency
```

java -cp target/HelloMaven-1.0-SNAPSHOT.jar com.example.App

```
skit@MANJUSHREENAVEEN: ~   skit@MANJUSHREENAVEEN: ·
-bash: syntax error near unexpected token `<'
skit@MANJUSHREENAVEEN:~/HelloMaven$ java -cp target/HelloMaven-1.0-SNAPSHOT.jar
 com.example.App
Hello World!
skit@MANJUSHREENAVEEN:~/HelloMaven$ gradle init
Starting a Gradle Daemon, 1 incompatible and 1 stopped Daemons could not be reu
sed, use --status for details
Reusing configuration cache.

Found a Maven build. Generate a Gradle build from this? (default: yes) [yes, no
]

Select build script DSL:
  1: Kotlin
  2: Groovy
Enter selection (default: Kotlin) [1..2] 2

Generate build using new APIs and behavior (some features may change in the nex
t minor release)? (default: no) [yes, no]


> Task :init
Maven to Gradle conversion is an incubating feature.
For more information, please refer to https://docs.gradle.org/8.14/userguide/mi
grating_from_maven.html in the Gradle documentation.

BUILD SUCCESSFUL in 1m 6s
1 actionable task: 1 executed
Configuration cache entry reused.
skit@MANJUSHREENAVEEN:~/HelloMaven$ tree
```

**expected output**

Hello World!

```
skit@MANJUSHREENAVEEN: ~   skit@MANJUSHREENAVEEN: ·
[INFO] Using 'UTF-8' encoding to copy filtered resources.
[INFO] skip non existing resourceDirectory /home/skit/HelloMaven/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.11.0:testCompile (default-testCompile) @ HelloMaven ---
[INFO] Changes detected - recompiling the module! :dependency
[INFO] Compiling 1 source file with javac [debug target 17] to target/test-classes
[INFO]
[INFO] --- maven-surefire-plugin:3.2.5:test (default-test) @ HelloMaven ---
[INFO] Tests are skipped.
[INFO]
[INFO] --- maven-jar-plugin:2.4:jar (default-jar) @ HelloMaven ---
[INFO] Building jar: /home/skit/HelloMaven/target/HelloMaven-1.0-SNAPSHOT.jar
[INFO] ------------------------------------------------------------------------
[INFO] BUILD SUCCESS
[INFO] ------------------------------------------------------------------------
[INFO] Total time:  4.457 s
[INFO] Finished at: 2026-02-24T06:10:01Z
[INFO] ------------------------------------------------------------------------
skit@MANJUSHREENAVEEN:~/HelloMaven$ java -cp <jar-file> <main-class>
-bash: syntax error near unexpected token `<'
skit@MANJUSHREENAVEEN:~/HelloMaven$ java -cp target/HelloMaven-1.0-SNAPSHOT.jar com.example.App
Hello World!
skit@MANJUSHREENAVEEN:~/HelloMaven$ gradle init
Starting a Gradle Daemon, 1 incompatible and 1 stopped Daemons could not be reused, use --status for details
Reusing configuration cache.

Found a Maven build. Generate a Gradle build from this? (default: yes) [yes, no]
```

Ensure you have successfully built and run the `HelloMaven` project using Maven before proceeding to Part B.

---

# Part B: Migrate the Application to Gradle (In-Place)

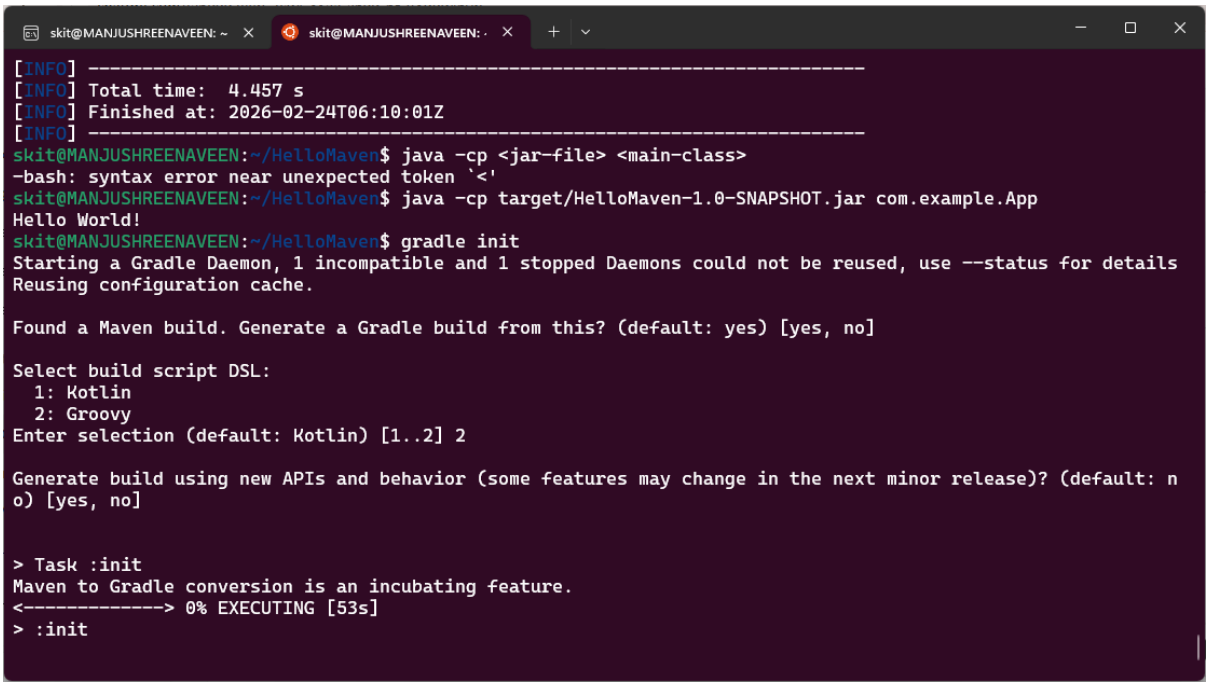
- **Step 1: Ensure You Are in the Maven Project Directory**

  Make sure your current directory in the terminal is the root of your Maven project (`HelloMaven`).

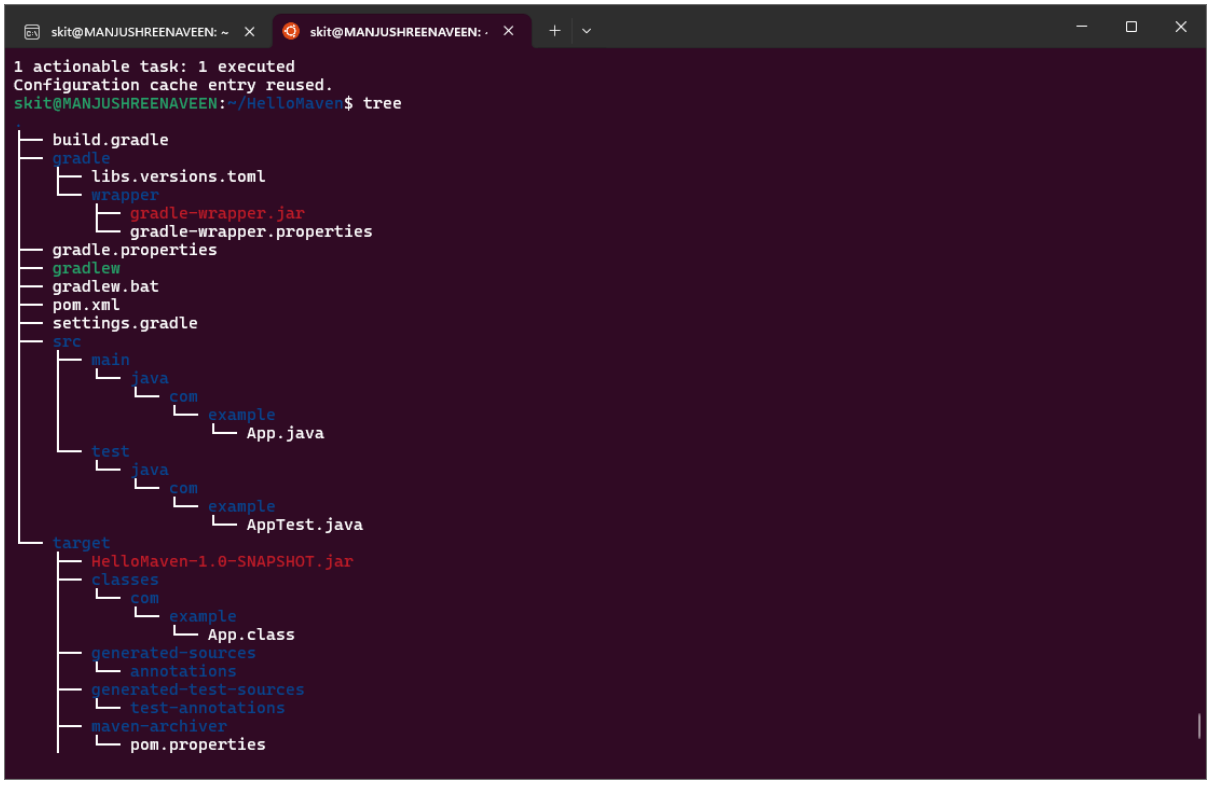
- **Step 2: Initialize Gradle in the Existing Maven Project**

  Run the `gradle init` command. Gradle will detect the existing project structure and attempt to generate appropriate build files. Select Groovy DSL.

  ```
  gradle init
  ```




- **Step 3: Explore the Updated Project Structure** After running `gradle init`, new files and directories will be added to your `HelloMaven` folder, including `gradlew`, `gradlew.bat`, `settings.gradle`, and potentially a `build.gradle` (or `build.gradle.kts`) file in the root or within a subfolder like `app` (depending on how `gradle init` interpreted your project). Use the `tree` command again to see the changes:
- `tree`




- **Step 4: Understand and Update the Generated Gradle Build Script** Gradle has generated a build script.

  Open the `build.gradle` file using command:

  ```
  gedit build.gradle
  ```

  Replace the entire content of the generated build script with the appropriate example below. The example here is configured to use JUnit 4 and correctly point the `application` plugin to the `com.example.App` class. Without `application` plugin `gradle run` will not work.

  **Groovy DSL (`build.gradle`):**

  ```
  /*

   * Gradle build file for HelloMaven - Experiment 4

   * Supports: gradle build & gradle run

   */



  plugins {

      id 'application'

      id 'java-library'
  ```

```
    id 'maven-publish'
}

group = 'com.example'
version = '1.0-SNAPSHOT'
description = 'HelloMaven'

java {
    sourceCompatibility = JavaVersion.VERSION_1_8
}

repositories {
    mavenLocal()
    mavenCentral()
}

dependencies {
    // JUnit 5 for testing
    testImplementation 'org.junit.jupiter:junit-jupiter:5.10.0'
}

application {
    // 🔴 MUST match your App.java package + class name
    mainClass = 'com.example.App'
}

test {
    useJUnitPlatform()
```

```
}


publishing {

    publications {

        maven(MavenPublication) {

            from components.java

        }

    }

}
```

Save the changes to the build script.

- **Step 5: Build the Project Using Gradle** From the root `HelloMaven` directory, use the Gradle command (`gradle`) to build the project. This will compile the code and run tests according to your Gradle build script.
- `gradle build`

  Look for a "BUILD SUCCESSFUL" message in the output.

- **Step 6: Run the Application Using Gradle**

  Run the app using this command:

  gradel run

```
skit@MANJUSHREENAVEEN: ~        skit@MANJUSHREENAVEEN: ·
skit@MANJUSHREENAVEEN:~/HelloMaven$ gedit build.gradle
skit@MANJUSHREENAVEEN:~/HelloMaven$ gradle clean build
Calculating task graph as no cached configuration is available for tasks: clean
 build

[Incubating] Problems report is available at: file:///home/skit/HelloMaven/buil
d/reports/problems/problems-report.html

Deprecated Gradle features were used in this build, making it incompatible with
 Gradle 9.0.

You can use '--warning-mode all' to show the individual deprecation warnings an
d determine if they come from your own scripts or plugins.

For more on this, please refer to https://docs.gradle.org/8.14/userguide/comman
d_line_interface.html#sec:command_line_warnings in the Gradle documentation.

BUILD SUCCESSFUL in 14s
8 actionable tasks: 8 executed
Configuration cache entry stored.
skit@MANJUSHREENAVEEN:~/HelloMaven$ gradle run
Calculating task graph as no cached configuration is available for tasks: run

> Task :run
Hello World!

BUILD SUCCESSFUL in 2s
2 actionable tasks: 1 executed, 1 up-to-date
Configuration cache entry stored.
skit@MANJUSHREENAVEEN:~/HelloMaven$
```